

Assignment No 4

Aim: Develop a Java program that performs various operations on arrays, including displaying elements, finding the maximum and minimum elements, calculating the sum and average of elements, and searching for a specific element within the array.

Code:

```
import java.util.Scanner;

public class ArrayOperations {

    public static void display(int[] arr){
        for (int i=0; i<arr.length; i++){
            System.out.println("Array Elements : " + arr[i]);
        }
    }

    public static void min(int[] arr) {
        int min = arr[0];
        for (int i = 1; i < arr.length; i++) {
            if (arr[i] < min) {
                min = arr[i];
            }
        }
        System.out.println("Minimum Element : " + min);
    }

    public static void max(int[] arr) {
        int max = arr[0];
        for (int i = 1; i < arr.length; i++) {
            if (arr[i] > max) {
                max = arr[i];
            }
        }
        System.out.println("Maximum Element : " + max);
    }

    public static void sumAverage(int[] arr) {
        double sum = 0;
        for (int i=0; i<arr.length; i++){
            sum = sum + arr[i];
        }
        System.out.println("Sum of Array Element : " + sum);
        double avg = sum / arr.length;
        System.out.println("Average of Array Element : " + avg);
    }

    public static void linearSearch(int[] arr, int key) {

        for (int i=0; i<arr.length; i++){
```

```

        if (arr[i] == key){
            System.out.println("Element is found at : " + i);
            return;
        }
    }
    System.out.println("Element is not found!");
}

public static void binarySearch(int[] arr, int key) {
    int low = 0;
    int high = arr.length - 1;

    while (low <= high) {
        int mid = (low + high) / 2;
        if (arr[mid] == key) {
            System.out.println("Element is found at index: " + mid);
            return;
        }
        else if (arr[mid] < key) {
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }
    System.out.println("Element is not Found");
}

```

```

public static void bubbleSort(int[] arr) {

    for (int i=0; i<arr.length-1; i++){
        for (int j=0; j<arr.length-i-1; j++){
            if (arr[j] < arr[j+1]){
                int temp = arr[j+1];
                arr[j+1] = arr[j];
                arr[j] = temp;
            }
        }
    }
}

```

```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    try {
        System.out.print("Enter the Number of Element : ");
        int n = sc.nextInt();
        int[] arr = new int[n];
    }
}

```

```

    for (int i = 0; i < arr.length; i++) {
        System.out.print("Enter the arr element " + i + " : ");
        arr[i] = sc.nextInt();
    }

    System.out.println();
    display(arr);
    min(arr);
    max(arr);
    sumAverage(arr);
    System.out.println("Enter the search element : ");
    int key = sc.nextInt();
    linearSearch(arr, key);
    binarySearch(arr, key);
    bubbleSort(arr);

} catch (Exception e) {
    System.out.println(e.getMessage());
}
}
}
}

```

Output:

```

Enter the Number of Element : 5
Enter the arr element 0 : 12
Enter the arr element 1 : 3
Enter the arr element 2 : 78
Enter the arr element 3 : 19
Enter the arr element 4 : 56

Array Elements : 12
Array Elements : 3
Array Elements : 78
Array Elements : 19
Array Elements : 56
Minimum Element : 3
Maximum Element : 78
Sum of Array Element : 168.0
Average of Array Element : 33.6
Enter the search element :
12
Element is found at : 0
Element is found at index: 0

Process finished with exit code 0

```